

A Pilot Growth-Lensing Test of the Reverse Universe Theory Bridge Using the E_G Statistic

Section 01 - Working Title, Abstract, Scope, and Claims Boundary

Draft status: Journal-style pilot/manuscript section for scientific circulation. This section introduces the paper cautiously and defines what is, and is not, being claimed.

Working Title

A Pilot Growth-Lensing Test of the Reverse Universe Theory Bridge Using the E_G Statistic

Alternative shorter title: **A Neutral-Lensing Bridge Test for Reverse Universe Theory in Pilot E_G Data.**

Abstract

Reverse Universe Theory (RUT) is a proposed geometric cosmology framework in which cosmic expansion, structure growth, lensing response, light propagation, and time-directionality are treated as measurable expressions of a shared transition structure. This paper does not attempt to test the full RUT framework. Instead, it isolates a reduced observational bridge model and compares it with LambdaCDM in background, growth, and growth-lensing diagnostics. The reduced bridge is encoded by $B(a) = 1/2[1 + \tanh((a - a_t)/\Delta a)]$, with a growth response $\mu(a) = 1 + \mu_0 B(a)$ and a lensing response $\Sigma(a) = 1 + \lambda_\Sigma \mu_0 B(a)$. Earlier background and growth tests showed that RUT is stable and mock-detectable, but that current background and joint growth data do not provide a decisive advantage over LambdaCDM. The strongest current result appears in the E_G growth-lensing lane. In a pilot diagonal E_G compilation, fixed neutral-lensing RUT, defined by $\lambda_\Sigma = 0$, improves the fit relative to LambdaCDM. The result survives one-measurement-per-family reduction, family-weighted averaging, and leave-one-point-out jackknife tests. In the hardened pilot packet, fixed neutral RUT gives Delta BIC approximately -2.108 in the full diagonal compilation, approximately -1.926 in the one-per-family subset, approximately -1.963 in the family-weighted subset, and wins 18/18 leave-one-point-out jackknife runs. We do not claim a detection. The analysis remains diagonal-only and not covariance-clean. The result is presented as a reproducible pilot signal and a target for independent, covariance-controlled testing.

Keywords

cosmology; modified gravity; growth of structure; gravitational lensing; E_G statistic; LambdaCDM; model comparison; Bayesian Information Criterion; Reverse Universe Theory

One-Sentence Claim

A fixed neutral-lensing RUT bridge shows a weak but robust pilot advantage over LambdaCDM in diagonal E_G growth-lensing comparisons, while full growth-lensing coupling is disfavored.

Claims Boundary

Allowed claim in this paper	Claim to avoid
RUT is testable through a reduced bridge model.	RUT is proven as a final theory of cosmology.
Fixed neutral-lensing RUT beats LambdaCDM in pilot diagonal E_G tests.	RUT has definitively replaced LambdaCDM.
The E_G result is robust under family-reduction and jackknife checks.	The E_G result is covariance-clean or confirmed.
The current signal motivates independent replication.	The current result is a discovery claim.

Planned Paper Structure

1. Introduction and conceptual motivation. 2. Minimal RUT bridge model. 3. Background and growth pretests. 4. The E_G observable and pilot dataset. 5. Fixed and diagnostic model cases. 6. Results. 7. Robustness and independence hardening. 8. Limitations. 9. Replication packet. 10. Conclusions.

Next Section to Draft

Section 02 should introduce RUT briefly: light, time, and measurable dimensional expressions as conceptual motivation, then immediately narrow the paper to the reduced bridge model tested here.

Section 02 - Conceptual Motivation

Light, Time, and Measurable Dimensional Expressions in the Reduced RUT Bridge Paper

2.1 Purpose of this section

Reverse Universe Theory (RUT) is motivated by a broader geometric hypothesis: cosmic expansion, gravitational structure, light propagation, and the directionality of time may not be independent phenomena, but different measurable expressions of a shared transition in the state of the universe. The purpose of this section is not to prove that broad framework. Instead, it gives the reader a concise conceptual motivation for why this paper tests RUT through growth and lensing, and why the reduced bridge model separates the growth response from the lensing response.

This distinction is important for the present manuscript. The full RUT framework contains additional claims about light, time, dimensional regimes, and field-state behavior. A journal-facing analysis should not require a reader to accept all of those claims. The present paper tests only a reduced observational structure: a bridge function $B(a)$, a growth response $\mu(a)$, a lensing response $\Sigma(a)$, and the E_G statistic as a growth-lensing diagnostic.

2.2 Light and time as operational quantities

In the broader RUT picture, light is treated not only as radiation, but also as an operational carrier of geometric information. Observational cosmology already uses light in this way: redshift, luminosity distance, angular diameter distance, lensing shear, and time delay all infer geometry from the propagation of light. RUT extends this intuition by asking whether the behavior of light also tracks a deeper transition between expansion-dominated and structure-dominated regimes.

Time is treated similarly. Rather than beginning with time as only an external parameter, RUT treats observable time-directionality as tied to expansion, compression, information flow, and field response. In this reduced paper, that idea is not inserted as metaphysics. It enters only through measurable functions of scale factor, especially through $B(a)$, which acts as an empirical bridge between regimes.

A conservative way to state the idea is:

light propagation + structure growth + lensing response = measurable probes of cosmic geometric transition

The paper therefore asks a narrow question: does a simple bridge model motivated by this transition improve growth-lensing consistency tests relative to Λ CDM?

2.3 Measurable dimensional expressions

RUT uses the phrase "dimensional expression" in an operational sense. A dimension or regime is not treated as physically meaningful simply because it can be named; it becomes meaningful only if it changes measurable behavior. For the purposes of this paper, a dimensional expression means a measurable response channel, such as expansion rate, growth rate, lensing response, clustering amplitude, or information propagation.

This journal paper therefore does not attempt to defend a complete dimensional ladder. Instead, it reduces the dimensional-regime idea to testable channels:

RUT idea	Reduced observable form in this paper
Expansion or background geometry	$H(z)$, distances, and $B(a)$

Structure growth	$\mu(a) = 1 + \mu_0 B(a)$
Lensing response	$\Sigma(a) = 1 + \lambda_{\Sigma} \mu_0 B(a)$
Growth-lensing consistency	E_G diagnostic
Regime transition	tanh bridge in scale factor

2.4 Why growth and lensing are allowed to separate

The Sprint C result makes this conceptual point important. A naive version of the model might force the same bridge response into both matter growth and lensing:

$$\mu(a) = 1 + \mu_0 B(a), \Sigma(a) = 1 + \mu_0 B(a)$$

This full-coupling option is simple, but the pilot E_G tests indicate that it is too strong. The E_G anchor and multi-point pilot comparison both disfavor a large positive lensing boost. The more successful form is neutral or mildly softened lensing:

$$\mu(a) = 1 + \mu_0 B(a), \Sigma(a) = 1 + \lambda_{\Sigma} \mu_0 B(a), \lambda_{\Sigma} \approx 0 \text{ to slightly negative}$$

This result is conceptually consistent with RUT because the theory does not require every response channel to move identically. A shared bridge can affect different measurable expressions differently. In plain terms: the universe may allow growth to respond to the bridge while lensing remains neutral, or even slightly softened.

2.5 Scope boundary for the paper

This manuscript does not claim that the full RUT framework has been established. It does not claim that light is proven to be the physical medium of time, nor that the full dimensional ladder has been empirically confirmed. Those ideas remain part of the broader theoretical motivation. The present paper makes a narrower and testable claim:

A reduced RUT bridge model with neutral lensing improves a pilot E_G growth-lensing comparison relative to LCDM across several diagonal robustness checks.

The central scientific value of the paper is therefore not the broadest philosophical interpretation of RUT. It is the fact that a specific reduced model produces a reproducible, falsifiable, data-facing result. This motivates a covariance-clean follow-up test and provides a concrete target for independent replication.

2.6 Transition to the formal model

The next section defines the reduced bridge model mathematically. The conceptual language of light, time, and dimensional expression will not be used as a substitute for evidence. From this point forward, the analysis is carried by the explicit functions $B(a)$, $\mu(a)$, $\Sigma(a)$, the pilot E_G data compilation, and the reported chi-square, AIC, BIC, and robustness tests.

Draft status: Section 02 prepared as part of a staged journal-style manuscript. Claims are intentionally limited to pilot-level, reproducible growth-lensing tests.

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Section 03 - Minimal RUT Bridge Model and Tested Cases

Purpose of this section: define the reduced RUT bridge used in the pilot tests, separate the growth and lensing response channels, and list the fixed and diagnostic cases compared against LambdaCDM.

3.1 Reduced bridge variable

The present paper does not test the full Reverse Universe Theory framework. It tests a reduced transition model designed to make the theory falsifiable against background, growth, and growth-lensing observables. The bridge variable is written as a smooth transition in scale factor:

$$B(a) = 1/2 [1 + \tanh((a - a_t)/\Delta a)]$$

where a is the scale factor, a_t is the transition scale, and Δa controls the width of the transition. In the Sprint A-C tests summarized in this paper, the reduced bridge uses:

$$a_t = 0.55, \Delta a = 0.08$$

This choice is not presented as a final theory parameterization. It is a controlled pilot form that allows RUT to be compared directly with LambdaCDM under standard model-selection statistics.

3.2 Background response

At the background level, the bridge modifies the effective dark-energy equation of state through a small radiation-like shift:

$$w_{DE}(a) = -1 + \beta B(a)/3$$

This form reflects the reduced RUT idea that the bridge moves the effective sector slightly toward a radiation-like response when the transition is active. The factor $1/3$ is not used here as a claim about literal photon domination; it is a phenomenological equation-of-state limit for the reduced test.

Sprint A background tests found that this background bridge is stable and mock-detectable, but real CC + DESI BAO background data do not prefer it over LambdaCDM. In the real-data background lane, β settled near a small value rather than producing a decisive model-selection advantage.

3.3 Growth response

For structure growth, the bridge is introduced through a phenomenological growth response:

$$\mu(a) = 1 + \mu_0 B(a)$$

Here $\mu(a)$ represents the effective response of nonrelativistic structure growth relative to the general-relativistic baseline. LambdaCDM corresponds to $\mu(a) = 1$. The Sprint B joint background + growth fit found a small positive growth response, approximately:

$$\mu_0 \simeq 0.173$$

This value is used as the fixed growth-channel input for the Sprint C E_G analysis. The paper treats it as a pilot value extracted from the preceding growth gate, not as a final universal constant.

3.4 Lensing response

The lensing channel is separated from the growth channel through:

$$\text{Sigma}(a) = 1 + \text{lambda_Sigma} \mu_0 B(a)$$

This equation is the key Sprint C model distinction. It allows growth and lensing to be related through the same bridge $B(a)$ without forcing them to respond identically. LambdaCDM has $\text{Sigma}(a) = 1$. Full growth-lensing coupling corresponds to $\text{lambda_Sigma} = 1$. Neutral lensing corresponds to $\text{lambda_Sigma} = 0$.

3.5 Tested model cases

Case	Definition	Role in paper
LambdaCDM	$\mu(a)=1$; $\text{Sigma}(a)=1$	Baseline comparison model.
RUT full coupling	$\mu(a)=1+\mu_0 B(a)$; $\text{Sigma}(a)=1+\mu_0 B(a)$	Fixed RUT case; strongly disfavored in pilot E_G tests.
RUT neutral lensing	$\mu(a)=1+\mu_0 B(a)$; $\text{Sigma}(a)=1$	Main fixed RUT case; no extra E_G-stage parameter.
RUT lambda-fit	$\mu(a)=1+\mu_0 B(a)$; $\text{Sigma}(a)=1+\text{lambda_Sigma} \mu_0 B(a)$	Diagnostic case; explores preferred lensing-channel response.

3.6 Why neutral lensing is the main fixed case

The neutral-lensing case is central because it is a fixed prediction once μ_0 is carried forward from Sprint B. It does not introduce an extra E_G-stage fitting parameter. Therefore, for this case, the BIC difference against LambdaCDM is simply the chi-square difference in the E_G comparison. This makes the result more conservative than a fitted lensing-channel model.

The pilot E_G results indicate that full coupling, $\text{lambda_Sigma} = 1$, pushes the lensing response too high. In contrast, neutral lensing, $\text{lambda_Sigma} = 0$, improves the E_G comparison relative to LambdaCDM across the full diagonal compilation and the hardened independence checks. The diagnostic fitted case generally prefers lambda_Sigma near zero to slightly negative, suggesting that growth and lensing may be related but not identical response channels.

3.7 Model-selection convention

The paper uses chi-square, AIC, and BIC as descriptive model-comparison statistics. For the fixed neutral-lensing RUT case, no additional E_G-stage parameter is counted. For the lambda-fit diagnostic case, one additional parameter is counted. A negative Delta BIC means that the RUT case improves the comparison relative to LambdaCDM under the stated penalty convention.

Because the pilot E_G compilation is diagonal-only, these statistics should not be interpreted as final discovery-level evidence. They are used to identify whether the reduced RUT bridge survives enough basic stress testing to justify covariance-clean follow-up work.

Transition to next section

The next section summarizes the background and growth pretests. Those pretests matter because they show that RUT does not simply win everywhere by construction: the background lane preferred LambdaCDM, the joint growth lane found only a mild positive μ_0 signal, and the E_G lane is where the first hardened pilot advantage appears.

Background and Growth Pretests: Sprint A and Sprint B

This section summarizes the pretests that motivated the growth-lensing analysis. The goal is not to claim that the background or growth-only sectors already establish RUT. Instead, these pretests define the reduced bridge, show where it does and does not improve on Lambda-CDM, and justify why the E_G statistic becomes the main target in the present paper.

4.1 Role of the Pretests

The minimal RUT bridge is tested in stages. Sprint A asks whether the bridge improves smooth background expansion observables such as $H(z)$ and BAO distance ratios. Sprint B asks whether the same reduced model leaves a measurable signature in the growth sector through $f\sigma_8$. These tests are necessary because a model that improves E_G only by breaking the background expansion or by requiring pathological growth behavior would not be scientifically useful.

The outcome is deliberately mixed. Sprint A and Sprint B do not provide a decisive real-data win over Lambda-CDM. Their value is that they constrain the allowed bridge behavior and identify the narrow region where a stable, data-facing RUT test remains plausible.

4.2 Sprint A: Background Bridge Test

Sprint A used a reduced background bridge with fixed transition location and width,

$$a_t = 0.55, \Delta a = 0.08$$

and a dark-energy equation-of-state deformation written schematically as

$$w_{DE}(a) = -1 + \beta B(a)/3.$$

The tested observables were cosmic-chronometer $H(z)$ points, DESI BAO distance ratios, and a broad sound-horizon prior. Mock-data checks showed that the pipeline behaves sensibly: when the mock data are Lambda-CDM-like, the bridge turns off or is penalized; when the mock data contain an injected RUT-like bridge, the reduced model can recover the injected signal.

On the real background data, however, the reduced bridge did not earn the extra parameter penalty. The best-fit bridge amplitude was small, with β approximately 0.1006, and the raw chi-square improvement was tiny. BIC continued to prefer Lambda-CDM.

Pretest	Dataset / Observable	Key result	Interpretation
Sprint A	CC $H(z)$ + DESI BAO + broad r_d prior	β approx 0.1006; $\Delta\chi^2$ approx -0.078; ΔBIC approx +3.71	Stable and mock-validated, but background data prefer Lambda-CDM.
Sprint B growth-only	Pilot $f\sigma_8$ compilation	Small positive growth modification allowed; not BIC-supported	Growth response is stable but weak.
Sprint B joint	CC + DESI BAO + r_d prior + $f\sigma_8$	μ_0 approx 0.173; $\Delta\chi^2$ approx -0.476; ΔBIC approx +3.63	Mild growth-sector improvement, but not enough to beat BIC.

4.3 Sprint B: Growth and Joint Growth-Background Test

Sprint B introduced a simple growth-sector response:

$$\mu(a) = 1 + \mu_0 B(a).$$

An unconstrained growth-only test initially allowed parameter escape, especially toward unrealistically low Ω_m . A broad background-informed prior on Ω_m removed that pathology and produced a healthier result: Ω_m remained near the background value and μ_0 settled inside the allowed range rather than hitting a boundary. This was an important numerical health check.

The joint test then fit the background and growth information together using CC, DESI BAO, the broad r_d prior, and pilot $f\sigma_8$ points. In that joint run, RUT improved the raw fit by approximately $\Delta\chi^2 = -0.476$, mostly through the $f\sigma_8$ sector, with a best-fit growth response near $\mu_0 = 0.173$. However, because this introduced an additional parameter, BIC still favored Lambda-CDM by approximately $\Delta\text{BIC} = +3.63$.

4.4 Why These Results Point Toward E_G

The pretests imply that the reduced RUT bridge is not primarily a background-expansion correction. In the background sector, the bridge is small and not BIC-supported. In the growth sector, a small positive μ_0 is allowed and mildly helpful, but still not decisive. This pattern suggests that the strongest test is not $H(z)$ alone or $f\sigma_8$ alone, but the relation between growth and lensing.

That relation is precisely the role of the E_G statistic. In a simplified phenomenological form, E_G compares matter clustering and velocity growth to the lensing potential. In the RUT bridge language, this makes E_G a direct probe of whether the growth channel $\mu(a)$ and the lensing channel $\Sigma(a)$ respond identically, separately, or oppositely to the bridge.

Reviewer-safe conclusion

Sprint A and Sprint B should not be presented as evidence that RUT has already defeated Lambda-CDM. They are pretests. Their scientific value is that they constrain the model, rule out overconfident background-only claims, and motivate a cleaner growth-lensing test. The main result of the present paper therefore begins only after these pretests: a fixed neutral-lensing E_G comparison in which $\Sigma(a)$ is not forced to follow $\mu(a)$.

Transition to Section 05

The next section defines the E_G observable, the pilot compilation used in the analysis, and the model-comparison procedure used to test Lambda-CDM, full-coupling RUT, neutral-lensing RUT, and diagnostic $\lambda\Sigma$ variants.

Section 05 - The E_G Observable and Pilot Dataset

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5.1 Purpose of the E_G Test

The background and growth pretests motivate a more targeted observable. Sprint A found that the reduced background bridge is stable and testable, but background expansion data alone do not favor the RUT bridge over LambdaCDM. Sprint B found a small positive growth-sector modifier, but the joint background-plus-growth fit still did not beat LambdaCDM after information-criterion penalties. The next logical target is therefore not the smooth expansion history alone, but the relationship between structure growth and gravitational lensing.

The E_G statistic is useful for this purpose because it combines information from galaxy clustering, redshift-space growth, and lensing. In the reduced RUT language, this makes E_G an ideal first observable for testing whether the bridge affects the growth channel and the lensing channel in the same way, in opposite ways, or not at all.

In this pilot paper, E_G is treated as a growth-lensing consistency diagnostic. The goal is not to claim a final measurement of modified gravity, but to test whether the reduced RUT bridge produces a more consistent pilot description of published E_G-style measurements than the LambdaCDM baseline.

5.2 Working E_G Approximation

For the present pilot comparison, the model prediction is constructed through the phenomenological approximation

$$E_G(z) \sim \Omega_{m0} * \Sigma(a) / f(z), \quad a = 1/(1+z)$$

where $f(z)$ is the linear growth rate and $\Sigma(a)$ controls the lensing response. This expression is not presented as the final RUT field-theory derivation of E_G. It is used as an operational bridge between the Sprint B growth calculation and the pilot E_G data. The approximation is sufficient for a first reproducibility packet because all compared cases are evaluated through the same diagnostic pipeline.

The LambdaCDM comparison uses $\Sigma(a)=1$ and general-relativistic growth. The RUT comparison uses the same bridge function developed in the earlier sections, with growth and lensing written as separate channels:

$$\mu(a) = 1 + \mu_0 B(a) \quad \Sigma(a) = 1 + \lambda_{\Sigma} * \mu_0 * B(a)$$

The Sprint B joint fit supplies the fixed growth amplitude $\mu_0 \sim 0.173$. In Sprint C, the central question becomes whether the lensing response prefers full coupling, neutral lensing, or a separate dual-channel response.

5.3 Tested Lensing Cases

The E_G analysis compares the following cases:

Case	Definition	Purpose
LambdaCDM	$\mu(a)=1, \Sigma(a)=1$	Benchmark model
RUT full coupling	$\lambda_{\Sigma}=1$	Tests whether lensing is boosted with growth
RUT neutral lensing	$\lambda_{\Sigma}=0$	Fixed no-extra-E_G-parameter RUT test
RUT lambda-fit	λ_{Σ} free	Diagnostic only; penalized by AIC/BIC

The neutral-lensing case is the most important fixed RUT test because it introduces no additional E_G-stage parameter. In that case, any BIC improvement relative to LambdaCDM is simply the chi-square improvement of a frozen prediction, rather than a reward from post-hoc fitting.

5.4 Pilot E_G Compilation

The pilot dataset contains 18 E_G-style measurements grouped into eight measurement families. The families used in the replication packet are Reyes, Blake, Amon, Alam, Pullen, Wenzl, Grimm, and Li_Xia. The compilation is intentionally broad: it is designed to test whether the RUT signal appears only in one measurement family or persists across a mixed growth-lensing dataset.

Family	Number of points	Role in pilot compilation
Reyes	1	Classic low-redshift anchor near $z \sim 0.32$
Blake	2	Historical E_G bins
Amon	3	Multiple lower-redshift bins
Alam	1	Single $z \sim 0.57$ comparison point
Pullen	1	Single $z \sim 0.57$ comparison point
Wenzl	2	Recent binned measurements
Grimm	4	Recent low-error redshift-bin set
Li_Xia	4	Recent binned lensing/growth comparison set

The full table of numerical values is included in the replication packet rather than reproduced in the main manuscript body. This keeps the paper readable while preserving reproducibility through the CSV files, verification script, and saved model-comparison tables.

5.5 Diagonal-Error Status and Claims Boundary

The pilot compilation uses diagonal errors. This is a major limitation. Several E_G-style measurements may share survey data, lensing maps, fiducial assumptions, or analysis pipelines. Therefore, the present study does not claim that the pilot compilation is a final independent covariance-clean dataset.

The purpose of the diagonal pilot analysis is narrower: to test whether the RUT neutral-lensing bridge is obviously disfavored, whether full coupling is too strong, and whether any apparent RUT advantage survives basic independence stress tests. Those stress tests include one-measurement-per-family reduction, family-weighted averaging, leave-one-family-out checks, and leave-one-point-out jackknife tests. The results of those tests are presented in the following sections.

Claims boundary: A diagonal E_G advantage is not a detection. A publication-grade claim requires covariance control, survey-overlap pruning, and an independently reproducible dataset. The present section defines the pilot data and model comparison framework only.

5.6 Reproducibility Assets

The replication packet accompanying this manuscript contains the Sprint A, Sprint B, Sprint C, Sprint C pilot E_G, and Sprint C-H hardening result packets. For the E_G analysis, the key reproducibility assets are the pilot E_G data CSV, the hardened subset comparison table, the leave-one-point-out jackknife table, and the verification script that checks the headline Delta BIC values.

This structure allows the reader to distinguish three levels of evidence: (i) the full diagonal compilation, (ii) family-reduced and family-weighted independence checks, and (iii) point-level jackknife robustness. The next two sections report the actual model-comparison outcomes.

Section 06 - Pilot E_G Results

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6.1 Overview of the Pilot Comparison

This section reports the first model-comparison result from the pilot E_G compilation defined in Section 05. The analysis compares LambdaCDM against three RUT bridge cases: full growth-lensing coupling, neutral lensing, and a diagnostic fitted lensing-channel coefficient. The central fixed RUT test is the neutral-lensing case, $\text{lambda_Sigma}=0$, because it introduces no additional E_G-stage fitted parameter.

The working model uses the Sprint B growth amplitude $\mu_0 \approx 0.173$ and evaluates the E_G diagnostic through the same computational pipeline for every case. Therefore, the result should be read as a relative comparison between model predictions under the same pilot assumptions, not as a final covariance-clean measurement of modified gravity.

6.2 Full Diagonal Compilation Result

On the full 18-point diagonal pilot compilation, the fixed neutral-lensing RUT case improves the raw chi-square relative to LambdaCDM:

$$\Delta \chi^2 = \chi^2_{\text{RUT_neutral}} - \chi^2_{\text{LCDM}} = -2.107592$$

Because the neutral-lensing case is treated as a fixed prediction in the E_G stage, it carries no additional E_G-stage parameter penalty. Thus, in this pilot comparison, the BIC difference is the same as the chi-square difference:

$$\Delta \text{BIC} = -2.107592$$

This is a weak-to-moderate pilot preference, not a decisive result. However, it is the first data-facing lane in the present testing sequence where the reduced RUT bridge improves over LambdaCDM under the BIC bookkeeping used in the replication packet.

Model	lambda_Sigma	Sigma0	chi2	k_EG	Delta BIC vs LCDM
LambdaCDM	-	-	20.289174	0	0.000000
RUT full coupling	1.000000	0.172759	56.318530	0	+36.029356
RUT neutral	0.000000	0.000000	18.181581	0	-2.107592
RUT anchor lambda	-0.426396	-0.073664	18.488367	0	-1.800807
RUT lambda-fit	-0.199992	-0.034550	17.092026	1	-0.306776

6.3 Failure of Full Growth-Lensing Coupling

The full-coupling case, $\text{lambda_Sigma}=1$, performs poorly in the pilot E_G comparison. It produces a chi-square of 56.318530, corresponding to $\Delta \text{BIC} = +36.029356$ relative to LambdaCDM. This is not a marginal failure; it is a strong indication that the E_G data do not want the RUT growth boost to be mirrored as an equal positive lensing boost.

This negative result is important. It prevents the model from being interpreted as a generic post-hoc adjustment that simply increases every channel. Instead, the data distinguish between possible RUT bridge implementations. In this pilot compilation, full growth-lensing coupling is too strong, while neutral lensing is preferred among the fixed cases.

6.4 Neutral Lensing as the Clean Fixed Case

The fixed neutral-lensing case sets $\Sigma(a)=1$ while retaining the Sprint B growth modifier through $\mu(a)=1+\mu_0 B(a)$. In practical terms, this means that the growth channel is allowed to receive the small RUT bridge correction inferred in the earlier growth analysis, while the lensing channel is not boosted by the same bridge.

$$\mu(a) = 1 + \mu_0 B(a) \quad \Sigma(a) = 1$$

This is the main result of the pilot E_G stage: a fixed neutral-lensing RUT bridge outperforms LambdaCDM in the full diagonal pilot compilation without adding a new E_G-stage parameter. The result is small, but it is cleaner than the lambda-fit case because it does not rely on tuning $\lambda\Sigma$ after seeing the E_G data.

6.5 Diagnostic Lambda Fit

Allowing $\lambda\Sigma$ to fit the full pilot E_G compilation lowers the raw chi-square further, from 18.181581 in the neutral fixed case to 17.092026 in the lambda-fit case. The best-fit value is

$$\lambda\Sigma \sim -0.199992, \quad \Sigma_0 \sim -0.034550$$

This suggests that the pilot E_G data prefer lensing to remain neutral or slightly softened relative to the growth channel. However, because $\lambda\Sigma$ is fitted directly to the E_G compilation, the lambda-fit case must be penalized by information criteria. After the BIC penalty, its advantage relative to LambdaCDM shrinks to $\Delta \text{BIC} \sim -0.306776$. This is weaker than the fixed neutral-lensing result.

Therefore, the lambda-fit result is interpreted as a diagnostic direction rather than the headline claim. It helps identify the preferred sign and scale of a possible dual-channel response, but the journal-safer result is the fixed neutral-lensing comparison.

6.6 Interpretation of the Pilot Result

The pilot E_G result points toward a specific RUT behavior: growth and lensing should not be forced to respond identically to the bridge. The data disfavor full coupling, mildly prefer neutral lensing as a fixed case, and allow a small negative lensing-channel coefficient as a diagnostic fit. In the language of the reduced model, the best current interpretation is

$$\mu_0 > 0, \quad \lambda\Sigma \sim 0 \text{ to slightly negative}$$

This aligns with the conceptual motivation introduced earlier: measurable dimensional or field-regime transitions may appear differently in growth and lensing observables. The pilot evidence does not prove that interpretation, but it provides a concrete observational target for the next, covariance-clean stage.

6.7 Conservative Claims Boundary

The correct claim from this section is narrow: in a diagonal pilot E_G compilation, fixed neutral-lensing RUT improves over LambdaCDM with $\Delta \text{BIC} \sim -2.108$. This is not a discovery-level result. It is a pilot advantage that justifies the hardening tests reported in the next section.

Conservative statement: The full-coupling RUT case is too strong, while the fixed neutral-lensing RUT case gives the best non-fitted bridge comparison against LambdaCDM in the full pilot E_G dataset. The result motivates robustness analysis, not a final detection claim.

Section 07 - Hardening and Robustness Checks

This section evaluates whether the pilot E_G advantage reported in Section 06 is fragile. The goal is not to claim a final detection, but to test the most immediate referee objection: that the result could be driven by one measurement family, one redshift cluster, or one unusually influential data point.

7.1 Robustness Question

The central robustness question is:

Does fixed neutral-lensing RUT still outperform Lambda-CDM when the E_G compilation is made stricter?

The fixed neutral-lensing case is used as the headline comparison because it adds no E_G-stage fitted parameter. In this case the model keeps the Sprint B growth modifier but sets the lensing response to neutral:

$$\mu(a) = 1 + \mu_0 B(a), \quad \Sigma(a) = 1.$$

This is contrasted with Lambda-CDM using the same pilot E_G observable and the same diagonal-error comparison. Since both Lambda-CDM and the fixed neutral RUT case are treated as fixed predictions at this stage, the BIC difference is equal to the chi-square difference for these subset comparisons.

7.2 Independence-Hardening Subsets

Three increasingly strict versions of the pilot E_G comparison were used. The first uses the full diagonal compilation. The second chooses one representative measurement from each survey family, selecting the minimum-error point within each family. The third replaces each family with a weighted-average point, producing one effective point per family.

These procedures do not replace a full covariance treatment. They are intermediate hardening tests designed to reduce obvious family overlap and identify whether the result depends on duplicated survey information.

Subset	n	Delta BIC: RUT neutral - LCDM	Interpretation
Full diagonal compilation	18	-2.108	Weak RUT-neutral preference
One-per-family minimum-error subset	8	-1.926	Survives family reduction
Family-weighted average subset	8	-1.963	Survives family compression

Table 7.1. Fixed neutral-lensing RUT remains preferred over Lambda-CDM after reducing obvious survey-family overlap. Negative Delta BIC favors RUT neutral.

7.3 Full Coupling Fails the Robustness Test

The full-coupling case, $\lambda_{\Sigma} = 1$, performs poorly in all hardened comparisons. This is important because it shows that the analysis is not simply selecting any RUT-like modification. A natural but overly strong version of the theory fails. The data prefer a separated growth-lensing response rather than identical growth and lensing activation.

In the full diagonal compilation, full coupling gives Delta BIC = +36.03 against Lambda-CDM; in the one-per-family subset it gives Delta BIC = +24.80; and in the family-weighted subset it gives Delta BIC = +35.13. These positive values are decisive evidence against full coupling within this pilot E_G setup.

7.4 Leave-One-Point-Out Jackknife

The strongest hardening check is the leave-one-point-out jackknife. Each E_G point is removed one at a time, and the fixed neutral-lensing RUT comparison is rerun on the remaining 17 points. This directly tests whether a single measurement is carrying the result.

The result is stable: fixed neutral-lensing RUT beats Lambda-CDM in all 18 leave-one-point-out runs.

$$\text{Neutral RUT win fraction} = 18 / 18 = 1.000$$

Worst jackknife Delta BIC = -1.266; best jackknife Delta BIC = -2.548.

The worst case occurs when one of the lower E_G points is removed, reducing the apparent advantage. Even then, fixed neutral RUT remains preferred. The best case occurs when removal of a single point strengthens the neutral RUT preference. The main conclusion is that no individual E_G measurement is necessary for the fixed neutral-lensing result to survive.

Jackknife metric	Value
Number of leave-one-point-out runs	18
Neutral RUT wins	18
Neutral RUT win fraction	1.000
Worst Delta BIC	-1.266
Best Delta BIC	-2.548

Table 7.2. Leave-one-point-out jackknife summary. Negative Delta BIC favors fixed neutral-lensing RUT over Lambda-CDM.

7.5 Fitted Lensing Channel as Diagnostic Only

The lambda_Sigma-fitted model is useful for diagnosing the preferred direction of the lensing response, but it is not used as the headline claim. In the full diagonal compilation, the best-fit value is approximately lambda_Sigma = -0.200. In the family-weighted subset, the best-fit value softens to approximately lambda_Sigma = -0.156. In the one-per-family minimum-error subset, the best-fit value is more negative, approximately lambda_Sigma = -0.560.

These values are not yet interpreted as physical measurements. They indicate only that the pilot E_G data prefer neutral or mildly softened lensing over full positive coupling. The fixed neutral case remains the cleaner claim because it does not introduce a fitted E_G-stage parameter.

7.6 Robustness Interpretation

Together, the hardening checks support a narrow and cautious conclusion: fixed neutral-lensing RUT has a weak but repeatable pilot advantage over Lambda-CDM in the E_G growth-lensing lane. This advantage survives the full diagonal compilation, one-per-family reduction, family-weighted compression, and all leave-one-point-out jackknife runs.

This does not establish RUT as a confirmed cosmological model. It establishes that the E_G lane is the strongest current target for independent replication. A publication-grade analysis must still include covariance treatment, survey-overlap auditing, and ideally a frozen holdout test.

Conservative Section 07 claim:

Fixed neutral-lensing RUT beats Lambda-CDM across the pilot E_G hardening checks, but the result remains pilot-level until covariance-clean and holdout analyses are completed.

Section 08 - Interpretation: Why Neutral Lensing Wins

The preceding sections show a specific pattern: RUT does not gain its strongest support from the background expansion tests, and full growth-lensing coupling is disfavored by the pilot E_G comparison. The strongest result occurs when the growth channel is allowed a small RUT bridge response while the lensing channel remains neutral. This section interprets that pattern without overstating it.

8.1 The Empirical Pattern

The tested growth-lensing bridge can be written as:

$$\mu(a) = 1 + \mu_0 B(a), \quad \Sigma(a) = 1 + \lambda_{\Sigma} \mu_0 B(a).$$

Sprint B supplied a small positive growth-channel amplitude, approximately $\mu_0 = 0.173$. Sprint C then asked how the lensing channel should respond. Three possibilities were compared: full positive coupling, neutral lensing, and a fitted diagnostic lensing channel.

The full-coupling case, $\lambda_{\Sigma} = 1$, boosts lensing by the same bridge response used in the growth channel. The pilot E_G data strongly disfavor that option. The neutral case, $\lambda_{\Sigma} = 0$, keeps $\Sigma(a) = 1$ and produces the most stable fixed-model improvement over Lambda-CDM. The fitted diagnostic case typically prefers λ_{Σ} near zero or mildly negative, not positive full coupling.

Case	Lensing response	Pilot interpretation
$\lambda_{\Sigma} = 1$	Full positive coupling	Too much E_G enhancement; strongly disfavored
$\lambda_{\Sigma} = 0$	Neutral lensing	Best fixed case; survives hardening checks
λ_{Σ} fitted	Diagnostic only	Prefers near-zero or mildly negative coupling

Table 8.1. Interpretation of the tested lensing-channel cases. The fixed neutral-lensing case is the paper's cleanest result because it adds no E_G-stage fitted parameter.

8.2 Physical Reading: Growth and Lensing Need Not Move Together

In many simple modified-gravity interpretations, a stronger gravitational response would be expected to affect both the motion of matter and the bending of light. The RUT pilot result suggests a more constrained possibility: the growth channel can show a small bridge response while lensing remains close to the Lambda-CDM value.

In plain terms, matter clustering may be slightly sensitive to the bridge, while light-deflection geometry may be protected, screened, or balanced by a compensating response. This is why the neutral-lensing case is not an arbitrary simplification. It is an empirically motivated boundary case: growth changes, but the lensing potential does not receive the same direct boost.

This interpretation is consistent with the broader RUT idea that different measurable dimensional expressions do not have to activate identically. A transition can appear in one observational channel while remaining suppressed in another. In the reduced model, this is represented by separating $\mu(a)$ from $\Sigma(a)$.

8.3 Why Full Coupling Fails

The full-coupling model assumes that the same bridge amplitude responsible for growth modification also increases lensing strength:

$$\lambda_{\Sigma} = 1 \Rightarrow \Sigma(a) = 1 + \mu_0 B(a).$$

This produces too large an E_G prediction in the pilot data. The result is not a mild failure; it is the worst of the tested RUT cases. This is scientifically useful. It shows that the analysis is not simply rewarding any RUT-shaped modification. The data are selecting a specific channel structure.

For this reason, the failure of full coupling should be treated as a constraint on RUT rather than a setback. It narrows the viable model space: RUT cannot simply raise both growth and lensing together. The theory must preserve a distinction between matter-growth response and lensing response.

8.4 Neutral Lensing as the Minimal Surviving Case

The neutral-lensing case is the minimal surviving case because it changes only the growth channel:

$$\mu(a) = 1 + \mu_0 B(a), \quad \Sigma(a) = 1.$$

This is important statistically. Since the model does not fit a new E_G -stage parameter, the BIC comparison is not being won by adding flexibility. The fixed neutral-lensing case outperforms Λ -CDM in the full diagonal compilation and remains preferred in the one-per-family subset, the family-weighted subset, and all leave-one-point-out jackknife checks.

The current result is therefore not that RUT has measured a new lensing coupling. The result is narrower and cleaner: the growth modifier inferred from Sprint B, when paired with neutral lensing, gives a repeatable weak advantage in pilot E_G data.

8.5 Relation to the Broader RUT Picture

The broader RUT framework treats light, time, and dimensional regimes as measurable expressions of an underlying transition. In that broader view, light propagation and matter growth are related, but not necessarily identical manifestations. The Sprint C result gives a data-facing version of that idea: the growth channel and the lensing channel should be modeled as coupled only through a controlled parameter, not forced to share the same amplitude.

The reduced paper does not attempt to prove the full RUT picture. It only uses the broader theory to motivate why the following separation is meaningful:

$$\begin{array}{ll} \text{Growth response: } \mu(a) = 1 + \mu_0 B(a) & \text{Lensing response: } \Sigma(a) = 1 + \lambda_\Sigma \mu_0 B(a) \end{array}$$

The pilot data currently prefer λ_Σ near zero over $\lambda_\Sigma = 1$. Thus the novel conceptual contribution is not merely that RUT predicts a bridge. It is that the bridge may express differently across observational channels.

8.6 Interpretation Boundary

The neutral-lensing result should not be interpreted as proof that lensing is exactly unmodified in nature. It should be interpreted as the strongest current fixed-case approximation under the pilot E_G data. A covariance-clean analysis could shift the preferred value of λ_Σ , and a future independent dataset could strengthen or weaken the current result.

The correct conclusion is therefore conditional and testable:

If the Sprint C result survives covariance-clean and holdout testing, RUT's best-supported growth-lensing prediction is not full coupling, but channel separation: modified growth with neutral or mildly softened lensing.

8.7 Section Summary

Section 08 interprets the main Sprint C pattern. The data disfavor full growth-lensing coupling, favor the fixed neutral-lensing case as the cleanest model, and suggest that future RUT tests should treat growth and lensing as related but non-identical channels. This gives a direct physical meaning to the hardened pilot result while preserving the paper's cautious claims boundary.

Section 09 - Limitations and Publication-Grade Requirements

This section defines what the present paper does not yet prove, why the Sprint C-H result should be treated as a hardened pilot result rather than a confirmed detection, and what would be required to make the analysis journal-ready in a stronger sense.

9.1 Current evidential status

The central result of the present analysis is that the fixed neutral-lensing RUT bridge improves a pilot E_G comparison relative to LCDM across several robustness checks. In the full diagonal compilation, the fixed neutral case gives Delta BIC approximately -2.108 relative to LCDM. In the one-measurement-per-family and family-weighted subsets, the same fixed neutral case remains favored, with Delta BIC approximately -1.926 and -1.963, respectively. In the leave-one-point-out jackknife, the neutral case wins in every run, with a worst-case Delta BIC of approximately -1.266.

These results are scientifically meaningful because the neutral-lensing case uses no additional E_G-stage fitted parameter. However, the evidence level remains weak-to-moderate. The correct interpretation is not that RUT has displaced LCDM, but that one specific RUT growth-lensing channel has produced a repeatable pilot advantage that justifies a more rigorous test.

9.2 Diagonal-error limitation

The main limitation is that the E_G comparison is diagonal-only. Each measurement is treated as if its uncertainty were independent of every other point. This is a practical first-pass approximation, but it is not the same as a covariance-clean likelihood.

Several E_G measurements may share survey data, lensing maps, galaxy samples, fiducial cosmology assumptions, redshift-space distortion inputs, or calibration systematics. If two points are partially correlated but are treated as independent, the apparent statistical strength can be overestimated. For this reason, the current analysis should be described as a pilot diagonal test.

9.3 Survey overlap and independence

The hardening pass partially addresses overlap by constructing one-measurement-per-family and family-weighted subsets. These tests reduce the chance that one measurement family dominates the result. The fixed neutral RUT advantage survives both reductions, which strengthens the pilot claim.

However, survey-family reduction is not a substitute for a true covariance analysis. A publication-grade version should identify which measurements share underlying data and either remove overlapping points, combine them using an explicitly stated prescription, or include a covariance matrix where one is available.

9.4 Fiducial cosmology and reconstruction dependence

Published E_G measurements are often extracted using assumed fiducial cosmologies, reconstruction pipelines, galaxy bias estimates, redshift-space distortion measurements, or lensing calibrations. A fully rigorous comparison should check whether the RUT prediction is being compared to measurements that were themselves derived under assumptions close to LCDM.

For the present pilot paper, this does not invalidate the result, but it limits the strength of the claim. The result should be framed as an observational consistency test using published E_G values, not as a reanalysis of the raw survey data.

9.5 Model-selection limitation

The use of AIC and BIC is helpful because it penalizes extra parameters, but AIC/BIC values are approximate model-selection tools. They are not replacements for a full likelihood analysis, Bayesian evidence calculation, or posterior predictive check. This matters especially for small pilot datasets.

The fixed neutral-lensing case is therefore the safest result to emphasize. Since $\lambda_{\text{Sigma}} = 0$ is fixed, the BIC improvement is simply the chi-square improvement. By contrast, the fitted λ_{Sigma} model is diagnostic: it helps identify the direction preferred by the data, but it should not be the headline result until tested against independent or covariance-clean data.

9.6 Publication-grade requirements

Requirement	Purpose	Clean-4 threshold
Covariance-clean E_G subset	Avoid overstating significance from correlated measurements.	Delta BIC remains below zero after covariance or strict independence cleaning.
Frozen-model holdout test	Prevent tuning to the same data used for evaluation.	Neutral RUT beats LCDM on data not used to shape the model.
Transparent replication packet	Allow independent verification of the numerical result.	One command or notebook reproduces all headline tables.
Survey-family audit	Document which points may overlap or share pipelines.	Every retained point has a clear inclusion rationale.
Independent follow-up observable	Test whether the signal is unique to the pilot E_G compilation.	Compatible result in lensing, Sigma, $P(k,z)$, or another growth-lensing observable.

9.7 Conservative claim allowed by the present analysis

The strongest defensible statement is:

Fixed neutral-lensing RUT improves a pilot diagonal E_G comparison relative to LCDM and survives one-per-family, family-weighted, and leave-one-point-out robustness checks.

The statement that should not be made is:

RUT is confirmed, or LCDM is ruled out.

The present results justify further testing. They do not yet justify a final cosmological claim.

9.8 Next analysis target

The immediate next step is a covariance and independence hardening pass. The model should be frozen before the next test:

$$\mu(a) = 1 + \mu_0 B(a), \text{Sigma}(a) = 1$$

with μ_0 taken from the background-plus-growth pretest rather than refit to the E_G data. A clean follow-up should then test this frozen prediction on a stricter E_G subset or an independent growth-lensing observable. If the fixed neutral-lensing result remains favored under that protocol, RUT would move from hardened pilot evidence toward a clean Tier-4 evidence claim in the E_G sector.

Section 10 - Conclusion and Next Work

This paper has not attempted to establish the full Reverse Universe Theory (RUT). Its narrower purpose was to test whether a reduced RUT bridge leaves a measurable signature in the relation between growth and lensing, using the E_G statistic as a pilot observable. The resulting conclusion is cautious but important: RUT does not yet have a confirmed observational victory over LCDM, but the neutral-lensing bridge has produced a hardened pilot advantage in the E_G lane.

10.1 Main conclusion

The central empirical result is that the fixed neutral-lensing RUT case improves the pilot E_G comparison relative to LCDM without introducing an extra E_G -stage parameter. In the full diagonal compilation, the result was:

$$\Delta \text{BIC}(\text{RUT neutral} - \text{LCDM}) \sim -2.108$$

This advantage remained when the dataset was reduced to one measurement per survey family and when each survey family was represented by a weighted average. It also survived the leave-one-point-out jackknife: fixed neutral RUT beat LCDM in 18 out of 18 jackknife runs. The worst jackknife case still gave a negative Delta BIC, approximately -1.266, and the best case reached approximately -2.548.

The result should therefore be described as a weak but hardened pilot advantage, not as a detection. The strength of the result is not only the numerical improvement; it is that the improvement persisted under independence-oriented stress tests designed to reduce obvious survey-family and single-point dependence.

10.2 Best current reduced RUT form

The tests favor the following reduced model as the cleanest data-facing version of the bridge:

$$\mu(a) = 1 + \mu_0 B(a)$$

$$\Sigma(a) = 1$$

where $B(a)$ is the bridge activation function and μ_0 is supplied by the growth-sector pretest. The more general diagnostic form remains useful:

$$\Sigma(a) = 1 + \lambda_{\Sigma} \mu_0 B(a)$$

but the present pilot data do not support full growth-lensing coupling. The $\lambda_{\Sigma} = 1$ case performs poorly. The fitted λ_{Σ} case tends toward neutral or mildly negative values, but because it introduces an additional parameter, it is best treated as diagnostic rather than the headline model.

10.3 Physical interpretation

The preferred result suggests that the RUT bridge, if physically meaningful, may affect structure growth without equivalently amplifying lensing. In plain terms, the data favor a small growth-channel response while the lensing channel remains neutral or slightly softened. This is consistent with the broader RUT idea that measurable dimensional regimes can separate into distinct observational channels, rather than all responding identically to one transition variable.

10.4 What would make the result journal-grade?

The next step is not to enlarge the theory. The next step is to make the current result harder to dismiss. A publication-grade version requires at least three improvements:

First, a covariance-clean E_G dataset. The current comparison uses diagonal errors. A referee will reasonably ask whether overlapping survey data, shared lensing maps, or common fiducial assumptions induce correlations. A cleaner analysis should either include covariance matrices where available or define a smaller, explicitly independent dataset.

Second, a frozen-model holdout test. The neutral-lensing case should be frozen before testing a separate dataset or subset. The preferred journal claim would be that the fixed model improves a held-out growth-lensing observable without tuning λ_{Sigma} or other E_G-stage parameters.

Third, an independent replication packet. The analysis should remain reproducible through public data files, model-definition scripts, summary tables, and a verification script. The present replication packet provides the foundation for that workflow, but the covariance-clean version should become the primary archival packet.

10.5 Recommended next work

The immediate next project should be a follow-up analysis titled something like: RUT Tier 4 Sprint C-H2 - Covariance-Clean E_G and Holdout Test. Its question should be narrow:

Does fixed neutral-lensing RUT still beat LCDM after covariance/independence cleaning and frozen-model holdout testing?

A positive answer would move RUT from a hardened pilot result toward a clean Tier-4 result in the E_G/growth-lensing sector. A negative answer would also be valuable, because it would identify the current pilot advantage as dataset-dependent rather than robust. Either outcome would strengthen the scientific discipline of the framework.

10.6 Final statement

The strongest defensible conclusion is therefore:

Fixed neutral-lensing RUT shows a weak but robust pilot advantage over LCDM in E_G growth-lensing comparisons, surviving family-reduced, family-weighted, and leave-one-point-out hardening checks. The result is promising enough for a journal-style pilot paper, but it requires covariance-clean replication before being treated as confirmed evidence.

Claims boundary: This section does not claim that RUT replaces LCDM. It states that the reduced neutral-lensing bridge has produced a reproducible pilot advantage in one observable lane and identifies the precise tests required to determine whether that advantage survives publication-grade scrutiny.

Section 11 - References and Replication Packet Notes

Purpose of this section. This section records the citation discipline, replication packet contents, verification procedure, and upload notes for the present manuscript. It is intended to make the pilot result easier to audit and harder to confuse with an unsupported theoretical claim.

11.1 Citation Status

The present manuscript should treat the current references as a working reference set. Before journal submission, every external literature entry should be checked against the original source and completed with author list, title, journal, year, volume, page or article number, DOI/arXiv identifier where available, and the exact data table or equation used.

The current paper relies on two kinds of sources:

- External scientific literature on the E_G statistic, modified gravity, growth, lensing, BAO, cosmic chronometers, and related cosmological model comparison.
- Internal replication artifacts produced during Sprints A, B, C, and C-H, including CSV result tables, packaged notebooks/results, and a verification script.

For the current draft, the external reference list should be treated as incomplete until the final bibliography pass. The replication packet, however, already provides an auditable record of the numerical tables used in the pilot result.

11.2 Working External Reference Categories

The final bibliography should include, at minimum, references for the following categories:

Category	Needed references / purpose
E_G definition and use	Original and follow-up papers defining E_G as a growth-lensing gravity diagnostic.
E_G measurements	Measurement papers corresponding to the pilot labels used in the packet: Reyes, Blake, Amon, Alam, Pullen, Wenzl, Grimm, and Li/Xia.
Λ CDM and cosmological parameters	Standard cosmology references for Λ CDM, BAO, cosmic chronometers, and matter growth.
Modified gravity language	References for $\mu(a)$, $\Sigma(a)$, gravitational slip, lensing modification, and growth-sector tests.
Information criteria	References or standard definitions for chi-square, AIC, BIC, and jackknife robustness logic.

11.3 Replication Packet Contents

The replication packet is designed to make the numerical claims traceable without requiring the reader to accept the manuscript narrative. The recommended repository folder is:

results/sprint_c_hardened_EG_replication_packet/

The packet should contain the following components:

Item	Role
README.md	Human-readable overview of the packet, claims boundary, and reproduction steps.
docs/ replication report PDF/MD	Short technical report summarizing Sprints A, B, C, and C-H.
source_packets/	Original Sprint A, B, C, C-pilot, and C-H zipped result packets.
data/extracted/	Extracted CSV tables used to verify the headline claims.

scripts/verify_results.py	Script that checks the headline Sprint C-H Delta BIC and jackknife claims from stored CSVs.
verification_output.txt	Expected output confirming that the packet passes internal verification.
MANIFEST.csv	File inventory for the packet.
CITATION.cff	Citation metadata for the packet release.

11.4 Headline Verification Targets

The verification script should reproduce the following headline checks from the stored Sprint C-H tables:

Check	Expected result
Full diagonal compilation	Fixed neutral-lensing RUT vs LCDM: Delta BIC approximately -2.108.
One-measurement-per-family subset	Fixed neutral-lensing RUT vs LCDM: Delta BIC approximately -1.926.
Family-weighted average subset	Fixed neutral-lensing RUT vs LCDM: Delta BIC approximately -1.963.
Leave-one-point-out jackknife	Neutral RUT win fraction = 1.0.
Worst jackknife case	Worst neutral Delta BIC remains negative, approximately -1.266.
Best jackknife case	Best neutral Delta BIC approximately -2.548.

11.5 Recommended Repository and OSF Upload Text

Recommended GitHub folder:

results/sprint_c_hardened_EG_replication_packet/

Recommended OSF folder:

RUT Results / Sprint C-H - E_G Independence Hardening

Recommended commit message:

Add Sprint C-H hardened E_G replication packet

Recommended short description:

Hardened pilot E_G replication packet for the Reverse Universe Theory neutral-lensing bridge test. Fixed neutral-lensing RUT is compared against LCDM across the full diagonal pilot compilation, one-per-family subset, family-weighted subset, and leave-one-point-out jackknife tests. This packet reports a promising pilot result and explicitly does not claim covariance-clean detection.

11.6 Claims Boundary for Submission

The manuscript should preserve the following language in any abstract, conclusion, repository summary, or cover letter:

- Use "pilot" and "diagonal-error" when describing the current E_G comparison.
- Use "hardened pilot evidence" rather than "confirmed detection."
- State that the result requires covariance-clean follow-up before any strong evidence claim.
- Treat fitted λ_{Sigma} as diagnostic; the headline model is fixed neutral lensing with $\lambda_{\text{Sigma}} = 0$.
- Avoid presenting RUT as a replacement for LCDM based on this paper alone. The narrower claim is an E_G growth-lensing advantage in the tested pilot framework.

11.7 Next Replication Milestone

The next version of the packet should add a covariance-clean or independence-clean E_G table, a frozen holdout test, and a full bibliography with checked source metadata. If fixed neutral-lensing RUT remains favored after those additions, the result would move from hardened pilot status toward a cleaner Tier-4-style evidence claim in the E_G/lensing sector.

Section 12 - Bibliography and Citation Map

Manuscript: A Pilot Growth-Lensing Test of the Reverse Universe Theory Bridge Using the E_G Statistic

12.1 Purpose of this section

This section adds the working bibliography for the journal-style manuscript and maps the references to the sections in which they should be cited. It is included as a manuscript-facing bibliography and as a practical citation checklist for the next submission edit. The replication packet remains a technical artifact rather than a peer-reviewed source.

12.2 E_G statistic - theory and definition

- [1] Zhang, P., Liguori, M., Bean, R., & Dodelson, S. (2007). Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity. *Physical Review Letters*, 99, 141302. arXiv:0704.1932. Role: original theoretical definition of E_G; cite when introducing E_G in Section 05.

12.3 E_G measurements - the eight pilot families

- [2] Reyes, R., Mandelbaum, R., Seljak, U., Baldauf, T., Gunn, J. E., Lombriser, L., & Smith, R. E. (2010). Confirmation of general relativity on large scales from weak lensing and galaxy velocities. *Nature*, 464, 256-258. arXiv:1003.2185. DOI:10.1038/nature08857. Measurement: E_G = 0.39 +/- 0.06 at z = 0.32. Role: classic low-redshift anchor and foundational observational E_G paper.
- [3] Blake, C., Joudaki, S., Heymans, C., Choi, A., Erben, T., Harnois-Deraps, J., Hildebrandt, H., Joachimi, B., Nakajima, R., van Waerbeke, L., & Viola, M. (2016). RCSLenS: Testing gravitational physics through the cross-correlation of weak lensing and large-scale structure. *MNRAS*, 456, 2806-2828. arXiv:1507.03086. DOI:10.1093/mnras/stv2875. Role: two historical E_G bins at z = 0.32 and z = 0.57.
- [4] Amon, A., Blake, C., Heymans, C., Leonard, C. D., Asgari, M., Bilicki, M., Choi, A., Erben, T., Glazebrook, K., Harnois-Deraps, J., Hildebrandt, H., Hoekstra, H., Joachimi, B., Joudaki, S., Kuijken, K., Lidman, C., Parkinson, D., Valentijn, E. A., & Wolf, C. (2018). KiDS+2dFlenS+GAMA: Testing the cosmological model with the E_G statistic. *MNRAS*, 479, 3422-3437. arXiv:1711.10999. DOI:10.1093/mnras/sty1624. Measurements: E_G at zbar = 0.267, 0.305, and 0.554. Role: three lower-redshift pilot entries.
- [5] Alam, S., et al. (BOSS Collaboration) (2017). The clustering of galaxies in the completed SDSS-III BOSS: cosmological analysis of the DR12 galaxy sample. *MNRAS*, 470, 2617-2652. arXiv:1607.03155. DOI:10.1093/mnras/stx721. Role: BOSS DR12 growth-rate input and z ~ 0.57 comparison context.
- [6] Pullen, A. R., Alam, S., He, S., & Ho, S. (2016). Constraining Gravity at the Largest Scales through CMB Lensing and Galaxy Velocities. *MNRAS*, 460, 4098-4108. arXiv:1511.04457. DOI:10.1093/mnras/stw1249. Measurement: E_G(z = 0.57) = 0.243 +/- 0.060(stat) +/- 0.013(sys). Role: CMB-lensing E_G using Planck x BOSS CMASS.
- [7] Wenzl, L., Bean, R., Chen, S.-F., Farren, G. S., Madhavacheril, M. S., Marques, G. A., Qu, F. J., Sehgal, N., Sherwin, B. D., & van Engelen, A. (2024). Constraining gravity with a new precision E_G estimator using Planck + SDSS BOSS. *Physical Review D*, 109, 083540. arXiv:2401.12971. DOI:10.1103/PhysRevD.109.083540. Role: recent high-precision harmonic-space E_G measurements.
- [8] Grimm, N., Bonvin, C., & Tutusaus, I. (2024). Testing General Relativity through the E_G Statistic Using the Weyl Potential and Galaxy Velocities. *Physical Review Letters*, 133, 211004. arXiv:2403.13709. DOI:10.1103/PhysRevLett.133.211004. Role: four-point recent low-error redshift-bin set.
- [9] Li, S. & Xia, J.-Q. (2025). Testing General Relativity using Large Scale Structures Photometric Redshift Surveys and Cosmic Microwave Background Lensing Effect. arXiv:2501.02852. Role: four recent DES MagLim + Planck CMB lensing binned measurements. Verify journal publication status before final submission.

12.4 Modified-gravity framework - mu(a), Sigma(a), and slip

- [10] Planck Collaboration - Aghanim, N. et al. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6. arXiv:1807.06209. DOI:10.1051/0004-6361/201833910. Role: LCDM baseline parameters.
- [11] Amendola, L. et al. (2018). Cosmology and fundamental physics with the Euclid satellite. *Living Reviews in Relativity*, 21, 2. arXiv:1606.00180. DOI:10.1007/s41114-017-0010-3. Role: mu(a), Sigma(a) modified-gravity parameterization.
- [12] Skara, F. & Perivolaropoulos, L. (2020). Tension of the E_G statistic and RSD data with Planck/LambdaCDM and implications for weakening gravity. *Physical Review D*, 101, 063521. arXiv:1911.10609. DOI:10.1103/PhysRevD.101.063521. Role: E_G and f sigma_8 tension background.

12.5 Background observables - cosmic chronometers and BAO

- [13] DESI Collaboration et al. (2024). DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations. arXiv:2404.03002. Role: DESI BAO measurements used in Sprint A. Verify journal publication status before final submission.
- [14] Moresco, M. et al. (2022). Unveiling the Universe with emerging cosmological probes. *Living Reviews in Relativity*, 25, 6. arXiv:2201.07241. DOI:10.1007/s41114-022-00040-z. Role: cosmic chronometer H(z) compilation.

12.6 Growth-rate data and model-selection statistics

- [15] Alam, S. et al. (BOSS Collaboration) (2017). Same as [5]. Additional role: f sigma_8 measurements for Sprint B growth.
- [16] Schwarz, G. E. (1978). Estimating the dimension of a model. *Annals of Statistics*, 6(2), 461-464. DOI:10.1214/aos/1176344136. Role: original BIC reference.
- [17] Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716-723. DOI:10.1109/TAC.1974.1100705. Role: original AIC reference.
- [18] Kass, R. E. & Raftery, A. E. (1995). Bayes Factors. *Journal of the American Statistical Association*, 90(430), 773-795. DOI:10.1080/01621459.1995.10476572. Role: interpreting BIC differences.

12.7 Replication artifact citation

[19] Van Dyk, M. (2026). RUT Tier 4 Sprint A+B+C Replication Packet - Background, Growth, and Hardened Pilot E_G Tests. GitHub: [results/sprint_c_hardened_EG_replication_packet/](#). OSF: RUT Results / Sprint C-H - E_G Independence Hardening. Version v1, 2026-05-13. Role: primary replication artifact. This is an archived technical artifact, not a peer-reviewed source.

12.8 Citation placement map for the next edit

Manuscript location	Primary citations to insert
Section 03	Planck [10], Euclid/modified gravity [11], AIC/BIC [16-18]
Section 04	DESI [13], Moresco [14], Alam/BOSS [5,15], replication artifact [19]
Section 05	Zhang [1], Reyes [2], Blake [3], Amon [4], Alam [5], Pullen [6], Wenzl [7], Grimm [8], Li & Xia [9]
Sections 06-07	AIC/BIC [16-18], replication artifact [19]
Section 11	Replication artifact [19] and source packet references

12.9 One-sentence summary for each E_G family

- Reyes (2010): Classic SDSS galaxy-galaxy lensing anchor at $z = 0.32$.
- Blake (2016): RCSLenS + BOSS two-bin measurement at $z = 0.32$ and $z = 0.57$.
- Amon (2018): KiDS+2dFLenS+GAMA three-bin measurement across $z = 0.27$ - 0.55 .
- Alam (2017): BOSS DR12 growth-rate input at $z \sim 0.57$.
- Pullen (2016): CMB-lensing E_G at $z = 0.57$ using Planck x CMASS.
- Wenzl (2024): High-precision harmonic-space E_G from Planck + SDSS BOSS.
- Grimm (2024): DES Weyl potential + spectroscopic velocities; four-redshift precision measurement.
- Li & Xia (2025): DES photometric + Planck CMB lensing; four-redshift binned measurement.

12.10 Submission checklist

- Verify Li & Xia (2025) journal publication status before final submission.
- Verify DESI 2024 journal publication status before final submission.
- Convert citations to target journal style: author-year for JCAP-like submissions or numbered citations for Physical Review D-like submissions.
- Do not cite the replication packet as peer-reviewed literature; cite it as a versioned technical archive.
- Keep the claims boundary intact: hardened pilot E_G advantage, not a confirmed detection.